ness. Table 4 presents the diversity measurement results. The experimental findings demonstrate that our method enhances diversity across all models. In some cases, we observed up to a threefold increase in the diversity of generated outputs.

Models	MTLD	TF-IDF	Lexical Entropy	Semantic Embedding	Self-BLEU
GPT-2	**41.36 ± 0.04**	0.87 ± 0.01	8.16 ± 0.09	0.55 ± 0.00	0.59 ± 0.11
GPT-2 + Text View	34.95 ± 1.14	**0.90 ± 0.01**	**8.35 ± 0.13**	**0.65 ± 0.01**	**0.65 ± 0.09**
Qwen-2.5	67.19 ± 0.14	0.80 ± 0.01	7.91 ± 0.05	0.22 ± 0.00	0.25 ± 0.09
Qwen-2.5 + Text View	67.64 ± 0.10	**0.86 ± 0.01**	**8.46 ± 0.07**	**0.38 ± 0.10**	**0.39 ± 0.10**
Qwen-2.5 + Image View	**70.28 ± 0.04**	0.84 ± 0.01	8.30 ± 0.07	0.30 ± 0.00	0.37 ± 0.11
LLama-3.2	52.45 ± 0.18	0.72 ± 0.01	7.41 ± 0.05	0.23 ± 0.00	0.18 ± 0.07
LLama-3.2 + Text View	54.23 ± 0.05	**0.84 ± 0.01**	**8.21 ± 0.09**	**0.43 ± 0.00**	**0.36 ± 0.11**
LLama-3.2 + Image View	**54.46** ± 0.07	0.76 ± 0.01	7.70 ± 0.06	0.31 ± 0.00	0.27 ± 0.07
DeepSeek-R1	52.51 ± 0.10	0.78 ± 0.01	7.64 ± 0.05	0.24 ± 0.00	0.24 ± 0.05
DeepSeek-R1 + Text View	**54.61 ± 0.21**	**0.85 ± 0.01**	**8.21 ± 0.08**	**0.39 ± 0.00**	**0.36 ± 0.09**
DeepSeek-R1 + Image View	54.36 ± 0.05	0.83 ± 0.01	8.07 ± 0.06	**0.39 ± 0.00**	0.34 ± 0.10
GPT-4O Mini	57.95 ± 0.02	0.63 ± 0.01	7.06 ± 0.02	0.11 ± 0.00	0.09 ± 0.05
GPT-4O Mini + Text View	**59.59 ± 0.11**	**0.81 ± 0.03**	**7.93 ± 0.03**	**0.33 ± 0.00**	**0.24 ± 0.11**
GPT-4O Mini + Image View	57.74 ± 2.59	0.79 ± 0.02	7.79 ± 0.03	0.30 ± 0.01	0.23 ± 0.08
GPT-4O	57.18 ± 0.71	0.66 ± 0.01	7.01 ± 0.14	0.11 ± 0.02	0.08 ± 0.04
GPT-4O + Text View	**58.25 ± 0.03**	0.79 ± 0.01	**7.86 ± 0.05**	**0.31 ± 0.00**	0.23 ± 0.08
GPT-4O + Image View	55.75 ± 1.49	**0.79 ± 0.01**	7.80 ± 0.08	**0.32 ± 0.01**	**0.26 ± 0.10**

Table 4: Mean ± standard deviation of diversity metrics across six different sample sizes ranging from 100 to 2000 per each prompt (10 prompts), with $max_length = 125$. Detailed results are provided in Appendix A.

Table 5 illustrates the novelty score of two well-known large language models GPT-4o and DeepSeek-R1 and impact of our method. By incorporating multi-view embeddings, we enriched the input representation, leading to the generation of more novel and diverse responses from these models. This analysis provides valuable insights into how multi-view embedding strategies influence novelty detection and enhance the creativity of LLM outputs. The results show that for GPT-4o, our approach in some cases led to around ninefold improvement in novelty when text or image view embeddings were applied. Similarly, for DeepSeek-R1—one of the most inherently creative models—our method resulted in approximately a twofold increase in the novelty score. Additional results for other models, evaluated solely using SBERT as the novelty detector, are presented in Table 14 in Appendix A.

The evaluation of answer correctness across different models is presented in Table 6, considering two aspects introduced in Section 3.2.3. We assess the correctness of generated answers based on their relevance to the input prompt, using GPT-4o as the correctness evaluator and evaluates the correctness of language models from an English language structure perspective with DeepSeekV3. The results indicate that all models achieve high correctness scores in this aspect. Figure 5 summarizes the performance of our proposed method (multi-view embedding) alongside standard model variants. We report three key metrics: diversity, novelty, and correctness. Diversity quantifies the variety of generated responses on scale of [0,1], novelty measures the proportion of novel content (expressed in percentages), and correctness reflects the relevance of the generated text to the given prompt. Overall, the data show that applying our method increases both diversity and novelty across various models. For instance, with Qwen, the baseline achieves a diversity of 0.41, novelty of 32.7%, and correctness of 93.77%. When we apply the text-view, diversity rises to 0.56 and novelty to 68.7%, albeit with a correctness drop to 76.6%. Similarly, the image-view offers 0.50 diversity and 57.2% novelty at a correctness of 82.5%. These patterns exemplify a trade-off: while the generated content becomes more varied and inventive, alignment with the prompt can decrease. To further evaluate whether these gains hold when only correct (i.e., relevant) answers are considered, we also compute diversity and novelty exclusively for the subset of correct responses. Even under this stricter measurement, the Qwen variants (text view and image view) maintain higher diversity and novelty than the baseline. This finding underscores that our method fosters creativity and variety without entirely sacrificing correctness..

	num_samples = 100		num_samples = 250		num_samples = 500
Model	GPT-4o	SBERT	GPT-4o	SBERT	SBERT
GPT-4o	10.60	5.4	7.28	3.52	2.9
GPT-4o + Text View	29.30	40.2	20.4	**32.08**	**24.48**
GPT-4o + Image View	**42.60**	**47.3**	**25.52**	30.44	22.54
DeepSeek-R1	27.40	43.6	18.6	36.24	31.82
DeepSeek-R1 + Text View	**46.20**	**75.3**	**38.22**	63.88	55.86
DeepSeek-R1 + Image View	40.60	73.7	27.56	**65.04**	**56.18**

Table 5: Results on percentage of novelty score across different models according to two novelty detectors, GPT-4o and SBERT. More results for other models exist in Table 14.

5 Conclusion

In this work, we present multi-view embedding, a model-agnostic approach that enriches the input prompt with diverse textual and visual sources to enhance the diversity and novelty of generated responses for both open-source and closed-source LLMs. Additionally, we introduce a framework to assess the diversity, novelty, and correctness of the generated outputs from LLMs. We validate both our method and framework using real-world datasets and $909k$ generated answers from various LLMs.

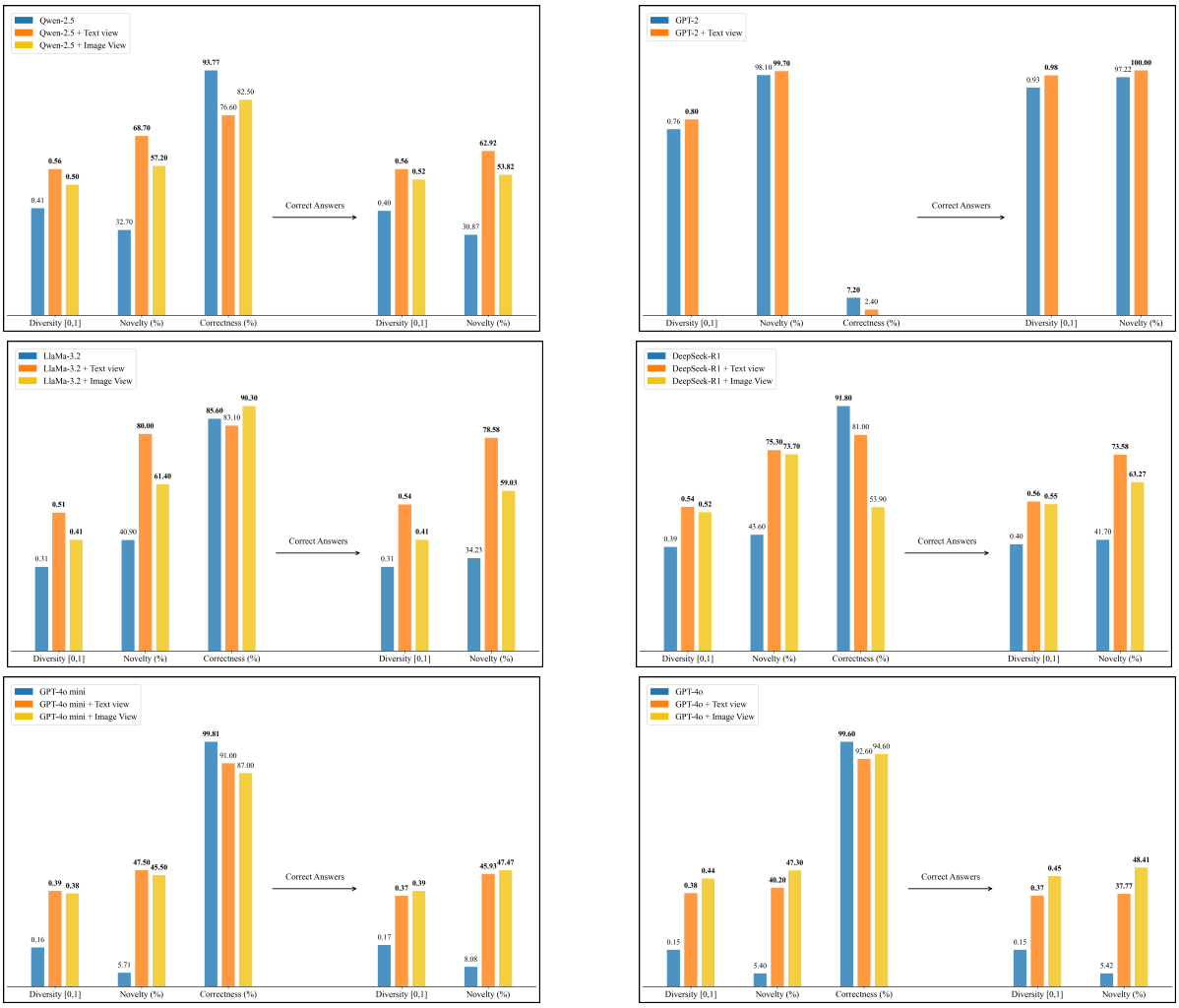

Figure 5: Diversity, Novelty, and Correctness (All answers) vs (Correct answers) For $num_samples = 100$. Diversity measure is $Self_BLUE$, Novelty measure is based on $SBERT$, and Correctness measure is based on $GPT-4o$.

6 Future Work

As demonstrated in this paper, incorporating textual or visual views enhances the output diversity and creativity of existing language models. In our future work, we aim to explore the impact of additional view types, such as videos, songs, maps, or even mathematical equations, as multi-view embeddings on model performance. Furthermore, we plan to extend our proposed framework to assess not only diversity, novelty, and correctness, but also effectiveness and usefulness, in order to provide a more comprehensive evaluation of the behavior of large language models.

	Correctness Measure	
Model	**Correctness (%)**	**Language Structure ([1,10])**
GPT-2	7.20	3.29
GPT-2 + Text View	2.40	2.40
Qwen	93.77	7.57
Qwen + Text View	76.60	7.52
Qwen + Image View	82.50	6.64
LlaMa-3.2	85.60	8.42
LlaMa-3.2 + Text View	83.10	7.38
LlaMa-3.2 + Image View	90.30	7.79
DeepSeek-R1	91.80	7.96
DeepSeek-R1 + Text View	81.00	7.15
DeepSeek-R1 + Image View	53.90	6.10
GPT-4o mini	99.81	8.70
GPT-4o mini + Text View	91.00	8.12
GPT-4o mini + Image View	87.00	7.71
GPT-4o	99.60	8.07
GPT-4o + Text View	92.60	8.05
GPT-4o + Image View	94.60	8.06

Table 6: Correctness (%) based on GPT-4o and language structure based on DeepSeekV3 ([1,10]). $num_sample = 100$ per each prompt (10 prompts).

7 Limitations

Despite the promising results, our method heavily depends on the quality and relevance of the views used during generation. If text or image views are not closely related to the input prompt, the model may produce outputs with lower correctness and reduced novelty. Additionally, generating these views—particularly in large numbers—can itself pose challenges. In our work, we use GPT-4o as a multi-view text generator; however, if many distinct views are requested, their diversity and novelty may diminish. This means that even if the underlying mechanism can foster creativity, oversaturating the process with a high volume of views can lead to repetitive or less innovative content. Finally, while our approach can enhance creativity and diversity for a moderate number of outputs, producing a large number of responses inevitably reduces both novelty and variation. These factors suggest that our method is most effective for situations where a bounded number of creative yet coherent outputs is desired, rather than an unbounded volume of responses.

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